SOFTWARE REQUIREMENTS SPECIFICATION

Vacation Home Insurance Management System

IEEE-aligned consolidated requirements baseline

Document Control	Value
Document ID	DOC-SRS-007
Version	2.0
Status	Final Revised SRS
Project Type	Full-Stack Insurance Capstone
Implementation Constraint	Approximately one week; enterprise-ready design baseline
Prepared For	Development, QA, Architecture, DevOps, Security and Stakeholder Sign-off
Release Date	26 September 2026

Document Purpose: This document is the authoritative implementation-ready baseline for the Vacation Home Insurance Management System. It incorporates the approved product model, payment workflow, claims assessment engine, document management, in-app notifications, caching, reporting, auditing, security controls, and updated traceability.

Document Revision History

Version	Date	Status	Summary
1.0	Baseline	Consolidated SRS Baseline	Original 5-coverage model and core workflows.
2.0	26-Sep-2026	Final Revised SRS	Replaced legacy model with three products; added payment, renewal, document management, notifications, caching, reporting, claims recommendation engine, Claims Manager approval, and expanded security/audit requirements.

Approval and Sign-off

Role	Name	Decision	Date
Product Owner			
Business Analyst			

Solution Architect			
Technical Lead			
QA Lead			

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1. Introduction

1.1 Purpose

This Software Requirements Specification consolidates the approved requirements, use cases, business rules, user journeys, architecture, domain model, database design, API design, security controls, quality attributes,

and traceability for the Vacation Home Insurance Management System. Version 2.0 supersedes the legacy five-coverage baseline and is the single authoritative reference for implementation and testing.

1.2 Scope

The system is a simplified digital platform through which a Customer can register, authenticate, maintain a profile, register vacation properties, compare approved insurance products, generate a risk-based Quote, complete a simulated payment, activate a one-year Policy, renew an eligible Policy, submit coverage-specific Claims with evidence, respond to evidence requests, and track Claim outcomes.

- A Claims Manager may self-register through a separate interface, upload one verification document, await Admin approval, review Claims, request further evidence, record structured assessments, receive a system recommendation, and make the final Claim decision.
- An Admin may approve or reject Claims Manager registrations, activate or deactivate user accounts, view operational records, activate or deactivate products, manage pricing and coverage limits, use the operational dashboard, and export approved reports.
- The system provides in-app notifications, local hybrid document storage, business audit logging, in-memory reference-data caching, controlled errors, and role-based authorization.

1.3 Business Context

Vacation properties differ from primary residences in occupancy, exposure, security, maintenance, and disaster risk. The platform demonstrates a complete insurance lifecycle rather than isolated CRUD functions.

```
Customer -> Property -> Product Selection -> Risk Assessment -> Premium Calculation ->
Quote -> Pending-Payment Policy -> Simulated Payment -> Active Policy ->
Coverage-Specific Claim -> Structured Review -> Recommendation -> Final Decision
```

1.4 Intended Audience

- Project evaluator
- Business analyst
- Product owner
- Solution architect
- Angular developer
- .NET developer
- Database engineer
- QA engineer
- DevOps engineer
- Security reviewer
- Future maintainer

1.5 Definitions

	Definition
Product	A customer-facing insurance package containing one or more fixed coverages.
Coverage	A specific protected peril or insured interest included in a Product and used for Claim eligibility.
Property Protection	Product containing Dwelling, Other Structures, Personal Property, and Theft Protection coverages.
Natural Disaster Protection	Product containing Flood, Earthquake, and Wildfire coverages.
Comprehensive Protection	Product containing all Property Protection and Natural Disaster Protection coverages.
Risk Assessment	Deterministic evaluation producing a score from 0 to 100 and Low, Medium, or High Risk Level.
Quote	A 30-day calculated offer containing product, coverage, risk, premium, and property snapshots.

Policy	A one-year insurance contract created from an applicable Quote and activated after successful simulated payment.
Claim Assessment	Claims Manager-entered review data used by the system to produce a non-binding recommendation.
Recommendation	System-generated Approve, Further Investigation, or Reject guidance; the Claims Manager remains the final decision maker.
Hybrid Document Storage	Metadata stored in SQL Server and physical files stored in controlled local server storage.
Business Audit Log	Persistent record of who performed a critical action, when, and the relevant old/new state.

1.6 Acronyms

Acronym	Meaning
API	Application Programming Interface
CM	Claims Manager
CSV	Comma-Separated Values
EF Core	Entity Framework Core
JWT	JSON Web Token
MVP	Minimum Viable Product
RBAC	Role-Based Access Control
REST	Representational State Transfer
RTM	Requirements Traceability Matrix
SRS	Software Requirements Specification
SQL	Structured Query Language
UTC	Coordinated Universal Time

1.7 References

- DOC-REQ-001, Requirements & Scope, Version 2.0.
- DOC-UCUS-002, Use Cases & User Stories.
- DOC-BR-003, Business Rules.
- DOC-ARCH-004, Architecture Document.
- DOC-DOMDB-005, Domain Model and Database Design.
- DOC-API-006, API Specification.
- Stakeholder clarification and requirements-discovery decision log completed before this release.

2. Overall Description

2.1 Product Perspective

```
Angular Frontend
   |
Ocelot API Gateway
   |-- Auth Service -> VacationHomeAuthDb
   |-- Customer Service -> VacationHomeCustomerDb
   |-- Insurance Service -> VacationHomeInsuranceDb + Local Document Storage +
In-Memory Cache
```

Exactly three business services are retained. Notifications and document management remain capabilities inside existing services and are not separate microservices. The Insurance Service contains Quote & Underwriting, Policy & Payment, and Claims modules.

2.2 Product Functions

- Customer registration, login, logout, forgot/reset/change password, and profile management.
- Vacation Property registration, viewing, and risk-relevant update.
- Product and included-coverage viewing.
- Risk assessment, premium calculation, Quote generation, history, and expiry.
- Pending-Payment Policy creation, simulated card payment, activation, cancellation, expiry, and manual renewal.
- Coverage-specific Claim submission, evidence upload, tracking, additional-evidence response, and outcome visibility.
- Claims Manager registration, verification-document upload, Admin approval, queue management, structured assessment, recommendation, and final decision.
- Admin dashboards, read-only operational visibility, user activation/deactivation, product activation/deactivation, pricing and coverage-limit configuration, reports, and CSV exports.
- In-app notifications, hybrid document storage, reference-data caching, business audit logging, error handling, and automated testing.

2.3 User Classes

Customer: Adult account holder who manages only their own profile, properties, Quotes, Policies, payments, Claims, documents, and notifications.

Claims Manager: Approved review user who accesses assigned/reviewable Claims, requests evidence, records assessments, receives recommendations, and makes final decisions.

Admin: Operational administrator who approves Claims Manager registrations, controls account eligibility, views system records read-only, manages products/pricing/limits, reports, and dashboard metrics. Admin cannot make Claim decisions.

2.4 Operating Environment

Layer	Approved Environment
Client	Modern browser; Angular SPA; responsive desktop-oriented UI.
Gateway	Ocelot API Gateway.
Services	.NET 8 ASP.NET Core Web APIs.
Persistence	SQL Server with EF Core and three service-owned databases.
Files	Controlled local server folders with SQL metadata.
Cache	ASP.NET Core in-memory cache for product, coverage, and reference data.
Communication	HTTPS REST/JSON; synchronous service calls for MVP.

2.5 Assumptions

- Pricing and risk parameters are capstone-only, deterministic, configurable values and do not represent insurer rates.
- The account holder and policy purchaser must be at least 18 years old; minor policyholder and guardian workflows are outside scope.
- Email is the login identifier and must be unique; phone is mandatory but need not be unique.
- Simulated card payment is used. No real funds are transferred.

- In-app notifications are mandatory for critical business events; real email integration is a future enhancement.
- No external property, payment, document, OCR, fraud, or insurer provider is required.
- Multiple concurrent authenticated sessions are allowed; a password change invalidates all existing sessions.

2.6 Constraints

- Approximately one-week implementation window.
- Angular, .NET 8, Ocelot, EF Core, and SQL Server are required.
- Exactly three business roles and three business services.
- No separate notification or document microservice.
- No Redis, message broker, distributed transaction, CQRS, Saga, Kubernetes, OCR, or production cloud orchestration.
- Hurricane Coverage and the legacy Fire, Theft, Natural Disaster, Personal Liability, and Personal Belongings customer-selectable catalog are removed.

2.7 Dependencies

- Quote generation depends on authoritative Customer Service property data and active Product/Pricing configuration.
- Policy activation depends on successful simulated payment.
- Claim submission depends on an owned Active Policy and a coverage included in that Policy.
- Claim decision depends on Under Review state and Claims Manager authority.
- Claims Manager login access depends on Admin approval.
- Document access depends on the parent record permission and document ownership context.

3. Business Overview

3.1 Vacation Home Insurance

The platform supports simplified vacation-home insurance with risk-based pricing and fixed customer-facing product bundles.

3.2 Approved Product Catalogue

Product	Included Coverages	Purchase Rule
Property Protection	Dwelling Coverage; Other Structures Coverage; Personal Property Coverage; Theft Protection	One product per policy.
Natural Disaster Protection	Flood Coverage; Earthquake Coverage; Wildfire Coverage	One product per policy.
Comprehensive Protection	All Property Protection and Natural Disaster Protection coverages	Mutually exclusive with separate Property Protection and Natural Disaster Protection policies for the same property.

A property may hold concurrent active Property Protection and Natural Disaster Protection policies. Comprehensive Protection replaces both and cannot overlap with either active product on the same property.

3.3 Risk Assessment

The system calculates a 0-100 Risk Score using Property Value, Property Age, Previous Claims Count, Disaster Exposure, Security System availability, and Occupancy Frequency. Risk Levels are Low (0-39), Medium (40-69), and High (70-100). The Customer cannot set or override the score.

Factor	Approved Points
Property Age	0-10 years: 5; 11-20: 10; 21-30: 15; over 30: 20
Previous Claims	0: 0; 1: 10; 2: 20; 3 or more: 30
Disaster Exposure	No: 0; Yes: 25
Security System	Yes: 0; No: 15
Occupancy Frequency	High: 0; Medium: 10; Low: 20
Property Value Band	Low: 0; Medium: 5; High: 10

The score is capped at 100. Property value bands are configurable approved reference data.

3.4 Premium

```
Base Premium = Property Value x Base Rate
Risk-Adjusted Premium = Base Premium x Risk Multiplier
Coverage Charges = Sum of charges for coverages included in selected Product
Final Premium = Risk-Adjusted Premium + Coverage Charges
```

The Customer sees Base Premium, Risk Adjustment, Coverage Charges, and Final Premium. Internal factor weights and detailed calculation configuration are not customer-editable. Existing Quotes preserve snapshots when pricing changes.

3.5 Coverage Limits

Coverage	Default Limit (% of Property Value)
Dwelling Coverage	80%
Other Structures Coverage	20%
Personal Property Coverage	15%
Theft Protection	10%
Flood Coverage	50%
Earthquake Coverage	60%
Wildfire Coverage	50%

Coverage limits are Admin-configurable for future Quotes and Policies. Existing Quote and Policy snapshots do not change retroactively.

3.6 Quote

A Quote is valid for 30 days, may coexist with other active Quotes for product comparison, and is retained with status Draft, Active, Expired, or Converted to Policy. A risk-relevant Property update invalidates affected active Quotes.

3.7 Policy

```
Pending Payment -> Active -> Expired
Pending Payment -> Cancelled
Active -> Cancelled
```

A Policy is one year in duration. Successful simulated payment immediately activates it; Effective Date is payment date and Expiry Date is system-calculated. Pending-Payment policies expire after 30 days.

Backdating and manual expiry changes are prohibited. Cancellation is customer initiated, reason mandatory, system validated, immediate, and final; no Admin approval is required.

3.8 Renewal

Manual renewal is available from 30 days before expiry through 30 days after expiry. Renewal recalculates risk and premium, permits product change subject to overlap rules, requires payment, and creates a new Policy Number. A cancelled Policy cannot be renewed, and only one active renewal process may exist at a time.

3.9 Claim

```
Submitted -> Under Review -> Awaiting Documents -> Under Review -> Approved / Rejected
Awaiting Documents -> Evidence Deadline Expired -> Under Review -> Approved / Rejected
```

Every Claim targets one specific coverage included in the Policy. Claim amount, incident date, description, severity, declaration, and at least one evidence file are required. Approved and Rejected are final. A rejected Claim may not be reopened; the Customer may submit a new Claim with new evidence.

3.10 Claim Recommendation

Assessment Factor	Weight
Coverage Match	40
Evidence Quality	20
Damage Severity Verification	20
Incident Validation	20

Scores of 80-100 recommend Approve, 60-79 recommend Further Investigation, and below 60 recommend Reject. Recommendation is advisory. The Claims Manager must record the final decision and a 20-1000 character reason.

4. Business Requirements

4.1 Customer Business Requirements

ID	Business Requirement	Priority	Status
BRQ-CUS-001	Register and authenticate as a Customer.	Must	Confirmed
BRQ-CUS-002	Manage own profile and change/reset password.	Must	Confirmed
BRQ-CUS-003	Register, view, and update owned vacation properties.	Must	Confirmed
BRQ-CUS-004	View active Products and included Coverages.	Must	Confirmed
BRQ-CUS-005	Generate, compare, and view own Quotes.	Must	Confirmed
BRQ-CUS-006	Create one Policy from an applicable Quote and complete simulated payment.	Must	Confirmed
BRQ-CUS-007	View, cancel, and renew own eligible Policies.	Must	Confirmed
BRQ-CUS-008	Submit a coverage-specific Claim with mandatory evidence.	Must	Confirmed
BRQ-CUS-009	Respond to additional evidence requests and track outcomes.	Must	Confirmed
BRQ-CUS-010	View and act on own in-app notifications.	Must	Confirmed

4.2 Claims Manager Business Requirements

ID	Business Requirement	Priority	Status
BRQ-CM-001	Self-register with one verification document and await Admin approval.	Must	Confirmed
BRQ-CM-002	Authenticate only when approved and active.	Must	Confirmed
BRQ-CM-003	View reviewable Claims and related approved information.	Must	Confirmed
BRQ-CM-004	Request additional evidence up to three times with a 15-day deadline.	Must	Confirmed
BRQ-CM-005	Upload investigation documents and record a structured assessment.	Must	Confirmed
BRQ-CM-006	Receive a non-binding system recommendation.	Must	Confirmed
BRQ-CM-007	Approve or reject an Under Review Claim with a mandatory reason.	Must	Confirmed
BRQ-CM-008	Complete final decision within seven days after investigation completion or evidence-deadline expiry.	Must	Confirmed

4.3 Admin Business Requirements

ID	Business Requirement	Priority	Status
BRQ-ADM-001	Access the protected operational dashboard.	Must	Confirmed
BRQ-ADM-002	Approve or reject Claims Manager registrations with mandatory rejection reason.	Must	Confirmed
BRQ-ADM-003	Activate or deactivate Customer and Claims Manager accounts.	Must	Confirmed
BRQ-ADM-004	View Customers, Properties, Policies, Claims, and Payments read-only.	Must	Confirmed
BRQ-ADM-005	Activate or deactivate approved Products; existing Policies remain valid.	Must	Confirmed
BRQ-ADM-006	Manage Base Rate, Risk Multipliers, Coverage Charges, and Coverage Limits.	Must	Confirmed
BRQ-ADM-007	View and export approved reports as CSV.	Must	Confirmed
BRQ-ADM-008	Admin shall not approve/reject Claims or alter active Policy/payment history.	Must	Confirmed

4.4 System Business Requirements

ID	Business Requirement	Priority	Status
BRQ-SYS-001	Enforce authentication, RBAC, ownership, and account eligibility.	Must	Confirmed
BRQ-SYS-002	Calculate deterministic Risk and Premium using versioned snapshots.	Must	Confirmed
BRQ-SYS-003	Enforce Product exclusivity, Quote validity, Policy, Payment, Renewal, and Claim lifecycles.	Must	Confirmed
BRQ-SYS-004	Persist business audit events and structured operational logs.	Must	Confirmed
BRQ-SYS-005	Provide secure hybrid document management.	Must	Confirmed
BRQ-SYS-006	Provide mandatory in-app notifications for critical events.	Must	Confirmed
BRQ-SYS-007	Cache Product, Coverage, and reference data for 30 minutes and invalidate immediately after Admin changes.	Must	Confirmed
BRQ-SYS-008	Provide controlled errors, concurrency checks, duplicate detection, reports, and CSV exports.	Must	Confirmed

5. Functional Requirements

5.1 Authentication and Account Management

FR-AUTH-001 Customer Registration

Attribute	Requirement
Description	Allow an adult person to register as a Customer using first name, last name, email, mandatory phone number, password, and confirmation.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Auth Service and Customer Service

Acceptance Criteria

- Email is the unique login identifier.
- Password has at least 8 characters, uppercase, lowercase, number, and special character.
- Account is created Active; email verification is not required.
- Age must be 18 or older before policy purchase.
- Sensitive values are not returned or logged.

FR-AUTH-002 Claims Manager Registration

Attribute	Requirement
Description	Allow a prospective Claims Manager to register using name, email, phone, years of experience, professional summary, password, and exactly one PDF/JPG/PNG verification document.
Actor	Claims Manager
Priority	Must
Status	Confirmed
Dependencies	Auth Service, Insurance document capability

Acceptance Criteria

- Account is Pending Approval and cannot access review functions.
- Admin can approve or reject.
- Rejection reason is mandatory and at least 20 characters.
- Rejected applicant may update and reapply.

FR-AUTH-003 Login and Session

Attribute	Requirement
Description	Authenticate Customer, approved Claims Manager, and Admin accounts and issue signed JWT tokens.
Actor	All
Priority	Must
Status	Confirmed
Dependencies	Eligible account

Acceptance Criteria

- Five failed attempts lock account for 15 minutes.
- Session expiration is 60 minutes.
- Authenticated state survives browser refresh.
- Multiple simultaneous sessions are allowed.

FR-AUTH-004 Forgot and Reset Password

Attribute	Requirement
Description	Generate a single-use reset token and allow a valid password reset.
Actor	All
Priority	Must
Status	Confirmed
Dependencies	Auth Service and notification capability

Acceptance Criteria

- Token is time-bound and cannot be reused.
- Reset obeys password policy.
- Existing sessions are invalidated after reset.

FR-AUTH-005 Change Password

Attribute	Requirement
Description	Allow authenticated user to change password using current password, new password, and confirmation.
Actor	All
Priority	Must
Status	Confirmed
Dependencies	Authenticated account

Acceptance Criteria

- Current password must be correct.
- All existing sessions are invalidated on success.
- Action is audited.

FR-AUTH-006 Logout

Attribute	Requirement
Description	End the active client session.
Actor	All
Priority	Must
Status	Confirmed
Dependencies	Authenticated context

Acceptance Criteria

- Subsequent protected access requires authentication.

FR-AUTH-007 Role and Account Authorization

Attribute	Requirement
Description	Enforce Customer, Claims Manager, and Admin permissions and active account status on the backend.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	JWT validation

Acceptance Criteria

- Customer self-role selection is prohibited.
- Admin Claim decision authority is denied.
- Non-owned Customer records are not disclosed.

5.2 Customer Profile

FR-CUST-001 View and Update Profile

Attribute	Requirement
Description	Allow Customer to view and update maintainable own profile fields.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Authentication

Acceptance Criteria

- Profile fields include date of birth, address lines, city, state, postal code, and country.
- Email is not editable through profile update.
- Customer cannot access another profile.

FR-CUST-002 Profile Completeness

Attribute	Requirement
Description	Require approved profile fields before Quote generation.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Customer Profile

Acceptance Criteria

- Incomplete profile blocks Quote generation with corrective guidance.

5.3 Vacation Property

FR-PROP-001 Register Property

Attribute	Requirement
Description	Allow Customer to register an owned vacation property.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Complete profile

Acceptance Criteria

- Fields include Property Name, Property Type, address, Property Value, Construction Year, Occupancy Frequency, Security System, Previous Claims Count, and Disaster Exposure.

- Property Value must be within configurable minimum 50,000 and maximum 10,000,000.
- Construction Year cannot be in future.
- Previous Claims Count is 0-10.

FR-PROP-002 View and Update Property

Attribute	Requirement
Description	Allow Customer to view and update only owned properties.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Authentication and ownership

Acceptance Criteria

- Property cannot be deleted or archived.
- Risk-relevant updates invalidate applicable active Quotes.
- Existing Policy snapshots are unchanged.

5.4 Product and Coverage Catalogue

FR-PROD-001 List Products

Attribute	Requirement
Description	Show only active Property Protection, Natural Disaster Protection, and Comprehensive Protection products with included Coverages.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Insurance Service cache

Acceptance Criteria

- Inactive Products are unavailable for new purchase and renewal.
- Existing Policies remain valid.

FR-PROD-002 Enforce Product Exclusivity

Attribute	Requirement
Description	Enforce one Product per Policy and prevent overlapping Comprehensive and component-product Policies for the same Property.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Policy data

Acceptance Criteria

- Property Protection plus Natural Disaster Protection may coexist.
- Comprehensive may not coexist with either on the same Property.

5.5 Risk, Pricing and Quote

FR-RISK-001 Calculate Risk

Attribute	Requirement
Description	Calculate 0-100 Risk Score and Low/Medium/High Risk Level using approved factors.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Complete Property data

Acceptance Criteria

- Customer cannot submit or override risk result.
- Identical inputs and configuration produce identical result.
- Calculation snapshot is traceable.

FR-PREM-001 Calculate Premium

Attribute	Requirement
Description	Calculate Base Premium, Risk-Adjusted Premium, Coverage Charges, and Final Premium.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Risk and active pricing configuration

Acceptance Criteria

- Coverage charges are summed from coverages included in selected Product.
- Customer receives summary breakdown.
- Pricing changes apply only to new Quotes.

FR-QUOTE-001 Generate Quote

Attribute	Requirement
Description	Generate an owned Quote for one selected Product and eligible Property.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Active Product, complete profile/property

Acceptance Criteria

- Quote is valid for 30 days.
- Quote stores Product, Coverage, Property, Risk, Pricing, and Premium snapshots.
- All Quotes are retained.
- Multiple active Quotes per Property are allowed.

FR-QUOTE-002 View and Compare Quotes

Attribute	Requirement
Description	Allow Customer to view own Quote history and compare active Quotes.
Actor	Customer

Priority	Must
Status	Confirmed
Dependencies	Ownership

Acceptance Criteria

- Statuses are Draft, Active, Expired, and Converted to Policy.

FR-QUOTE-003 Invalidate Stale Quote

Attribute	Requirement
Description	Invalidate active Quote when a material Property input changes or Product is no longer purchasable.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Quote and Property

Acceptance Criteria

- Expired, converted, or invalid Quotes cannot create a Policy.

5.6 Policy, Payment, Cancellation and Renewal

FR-POL-001 Create Pending-Payment Policy

Attribute	Requirement
Description	Create one Pending Payment Policy from an applicable owned Quote.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Valid Quote

Acceptance Criteria

- Policy has unique POL-YYYY-NNNNN number.
- One Quote converts at most once.
- Policy starts Pending Payment.

FR-PAY-001 Simulated Card Payment

Attribute	Requirement
Description	Provide mock Credit/Debit Card payment for a Pending Payment Policy.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Pending Payment Policy

Acceptance Criteria

- Fields: cardholder name, card number, expiry date, CVV, amount.
- No real card data or funds are transmitted.

- Every attempt is stored with PAY-YYYY-NNNNN number.
- Maximum five attempts per payment process.

FR-PAY-002 Activate Policy

Attribute	Requirement
Description	Immediately activate Policy after successful simulated payment.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Successful payment

Acceptance Criteria

- Effective Date is payment date.
- Expiry Date is system-calculated one year later.
- Activation notification and audit event are created.

FR-POL-002 Expire Pending Payment

Attribute	Requirement
Description	Expire unpaid Pending Payment Policy after 30 days.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Scheduled evaluation

Acceptance Criteria

- Expired Pending Payment Policy cannot be paid.

FR-POL-003 Cancel Policy

Attribute	Requirement
Description	Allow Customer to cancel own Pending Payment or Active Policy after supplying a mandatory reason.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Eligible owned Policy

Acceptance Criteria

- No Admin approval is required.
- Cancellation is system validated, immediate, final, audited, and notified.

FR-POL-004 Renew Policy

Attribute	Requirement
Description	Allow eligible Policy renewal 30 days before expiry through 30 days after expiry.
Actor	Customer
Priority	Must

Status	Confirmed
Dependencies	Eligible non-cancelled Policy

Acceptance Criteria

- Risk and Premium are recalculated.
- Product may change subject to exclusivity rules.
- New Quote, payment, Policy, and Policy Number are created.
- Only one active renewal process is allowed.

5.7 Claims and Documents

FR-CLM-001 Submit Claim

Attribute	Requirement
Description	Allow Customer to submit Claim against a specific included Coverage of an owned Active Policy.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Active Policy and documents

Acceptance Criteria

- Required: Incident Date, Claim Amount, Description, Damage Severity, 1-10 evidence files, and declaration.
- Incident Date must be within Policy coverage period and not future.
- Claim Amount must be greater than zero and within Coverage Limit.
- Possible duplicates are flagged but not automatically blocked.
- Claim number format is CLM-YYYY-NNNNN.

FR-DOC-001 Upload Claim Evidence

Attribute	Requirement
Description	Upload PDF, JPG, or PNG files using hybrid storage.
Actor	Customer
Priority	Must
Status	Confirmed
Dependencies	Claim context

Acceptance Criteria

- Each file is at most 10 MB.
- Minimum one and maximum ten Claim files.
- Customer cannot delete uploaded evidence.
- File type, size, name, ownership, and path are validated.

FR-CLM-002 Begin Review

Attribute	Requirement
Description	Move Submitted Claim to Under Review through Claims Manager action.
Actor	Claims Manager

Priority	Must
Status	Confirmed
Dependencies	Approved Claims Manager

Acceptance Criteria

- Review start is audited and Customer is notified.

FR-CLM-003 Request Additional Evidence

Attribute	Requirement
Description	Request additional evidence up to three times.
Actor	Claims Manager
Priority	Must
Status	Confirmed
Dependencies	Under Review Claim

Acceptance Criteria

- Each request has reason, requested items, and 15-day deadline.
- Claim becomes Awaiting Documents.
- Customer is notified.
- After deadline, status becomes Evidence Deadline Expired; no automatic rejection occurs.

FR-CLM-004 Upload Investigation Document

Attribute	Requirement
Description	Allow Claims Manager to upload PDF/JPG/PNG investigation documents.
Actor	Claims Manager
Priority	Must
Status	Confirmed
Dependencies	Review context

Acceptance Criteria

- Access and download are audited.

FR-CLM-005 Record Assessment and Recommendation

Attribute	Requirement
Description	Record Coverage Match, Evidence Quality, Damage Severity Verification, Incident Validation, and Estimated Loss; then calculate recommendation.
Actor	Claims Manager/System
Priority	Must
Status	Confirmed
Dependencies	Under Review Claim

Acceptance Criteria

- Coverage Match values: Covered, Not Covered, Uncertain.
- Evidence Quality: Low, Medium, High.
- Incident Validation: Valid, Requires Further Investigation, Suspicious.

- Recommendation does not finalize the Claim.

FR-CLM-006 Decide Claim

Attribute	Requirement
Description	Allow Claims Manager to approve or reject an Under Review Claim.
Actor	Claims Manager
Priority	Must
Status	Confirmed
Dependencies	Completed assessment

Acceptance Criteria

- Decision reason is 20-1000 characters.
- Approved and Rejected are final.
- Decision is expected within seven days after investigation completion or evidence deadline expiry.
- Customer is notified.

5.8 Notifications

FR-NTF-001 Create In-App Notification

Attribute	Requirement
Description	Create mandatory in-app notifications for approved business events.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Notification table

Acceptance Criteria

- Notification includes title, message, type, record link, created time, and read status.
- Notification can deep-link to related record.
- Critical Claim and Policy notifications cannot be disabled.

FR-NTF-002 Notification History

Attribute	Requirement
Description	Allow users to view notification history and mark notifications read.
Actor	All
Priority	Must
Status	Confirmed
Dependencies	Authentication

Acceptance Criteria

- Notifications are retained as business history and not automatically deleted.

5.9 Administration and Reporting

FR-ADM-001 Claims Manager Approval

Attribute	Requirement
Description	Allow Admin to review verification information and approve or reject a Claims Manager registration.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Pending Approval account

Acceptance Criteria

- Rejection requires reason.
- Applicant is notified.
- Action is audited.

FR-ADM-002 Account Eligibility Management

Attribute	Requirement
Description	Allow Admin to activate/deactivate Customer and Claims Manager accounts.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Admin authorization

Acceptance Criteria

- Admin cannot edit Customer profile or insurance history.

FR-ADM-003 Product Status Management

Attribute	Requirement
Description	Allow Admin to activate or deactivate approved Products.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Product catalogue

Acceptance Criteria

- Admin cannot create or delete Products.
- Existing Policies and Claims remain valid.
- Cache is invalidated immediately.

FR-ADM-004 Pricing and Limit Configuration

Attribute	Requirement
Description	Allow Admin to maintain Base Rate, Risk Multipliers, Coverage Charges, and Coverage Limit percentages.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Admin authorization

Acceptance Criteria

- Values must be non-negative and valid.
- Changes are effective only for new Quotes.
- Old/new values are audited and cache invalidated.

FR-ADM-005 Operational Dashboard

Attribute	Requirement
Description	Display approved system metrics.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Reporting queries

Acceptance Criteria

- Metrics: customers, active customers, pending CM approvals, active policies, pending payments, active claims, active/inactive products, and Claim approval rate.

FR-RPT-001 Reports and CSV Export

Attribute	Requirement
Description	Provide Customer, Property, Policy, Claim, Payment, and Product reports with filters and CSV export.
Actor	Admin
Priority	Must
Status	Confirmed
Dependencies	Read-only operational data

Acceptance Criteria

- Export is authorized and audited.
- No secrets or document binary content are exported.

5.10 Caching

FR-CACHE-001 Reference Data Cache

Attribute	Requirement
Description	Cache Product catalogue, Coverage catalogue, pricing/reference data in ASP.NET Core in-memory cache for 30 minutes.
Actor	System
Priority	Must
Status	Confirmed
Dependencies	Insurance Service

Acceptance Criteria

- Next request reloads data after expiration.
- Admin activation/deactivation and configuration updates invalidate related cache keys immediately.
- Redis is not required.

6. User Stories

US-AUTH-001 As a Customer, I want to register and log in so that I can use my own insurance workspace.

US-CM-001 As a prospective Claims Manager, I want to submit professional information and verification evidence so that an Admin can approve my access.

US-PROP-001 As a Customer, I want to register my vacation property so that it can be evaluated.

US-PROD-001 As a Customer, I want to see products and included coverages so that I understand what I am buying.

US-QUOTE-001 As a Customer, I want a transparent risk-based Quote so that I can compare available products.

US-PAY-001 As a Customer, I want to complete a simulated payment so that my Policy becomes Active.

US-REN-001 As a Customer, I want to renew an eligible Policy and receive a newly priced Policy.

US-CLM-001 As a Customer, I want to submit a coverage-specific Claim with evidence so that it can be reviewed.

US-CM-002 As a Claims Manager, I want to request evidence, assess the Claim, and receive a recommendation so that I can make a responsible final decision.

US-ADM-001 As an Admin, I want to approve Claims Managers and manage operational configuration without altering insurance decisions.

US-NTF-001 As a user, I want relevant notifications linked to records so that I can act promptly.

US-RPT-001 As an Admin, I want reports and CSV exports so that I can monitor operations.

7. Use Cases

UC-AUTH-001 Register Customer

Attribute	Value
Actor	Customer
Goal	Create a Customer account.
Preconditions	Valid registration data is available.
Postconditions	Active Customer account and Customer context are created, or request is rejected.

Main Flow

1. Customer opens registration.
2. Customer enters required identity, contact, and password fields.
3. System validates email uniqueness and password policy.
4. Auth Service creates a Customer-role account.
5. Customer Service creates the Customer context.
6. System records audit event and confirms registration.

Exception Flow

- Duplicate email.
- Validation failure.
- Partial onboarding is logged and recoverable.

UC-CM-001 Register and Approve Claims Manager

Attribute	Value

Actor	Claims Manager, Admin
Goal	Obtain authorized review access.
Preconditions	Applicant supplies professional details and one valid document.
Postconditions	Account becomes Active or Rejected.

Main Flow

7. Applicant submits registration and verification document.
8. System creates Pending Approval account.
9. Admin opens pending request.
10. Admin reviews profile and document.
11. Admin approves or rejects with reason.
12. System updates status, audits action, and notifies applicant.

Alternative Flow

- Rejected applicant updates details and reapplies.

UC-QUOTE-001 Generate Quote

Attribute	Value
Actor	Customer
Goal	Obtain a 30-day risk-based offer.
Preconditions	Customer owns eligible property and selects active Product.
Postconditions	Quote is stored or rejected.

Main Flow

13. Customer selects Property and Product.
14. System validates profile, property, product, overlap, and data completeness.
15. Insurance Service obtains authoritative Property data.
16. System calculates Risk Score and Level.
17. System applies pricing configuration and coverage charges.
18. System stores Quote snapshots and expiry.
19. System returns premium breakdown and included coverages.

UC-POL-001 Purchase Policy

Attribute	Value
Actor	Customer
Goal	Create and activate Policy from Quote.
Preconditions	Applicable owned Quote exists.
Postconditions	Active Policy or recorded failed payment state.

Main Flow

20. Customer converts Quote to Pending Payment Policy.
21. System assigns Policy Number.
22. Customer opens mock payment screen.
23. System validates simulated card fields and stores attempt.
24. On success, Policy becomes Active for one year.
25. System creates notifications and audit events.

Alternative Flow

- Failed payment leaves Policy Pending Payment and permits retry up to five attempts.

UC-CLM-001 Submit Claim

Attribute	Value
Actor	Customer
Goal	Submit coverage-specific Claim.
Preconditions	Owned Active Policy and required evidence exist.
Postconditions	Submitted Claim created or rejected.

Main Flow

26. Customer selects Policy and included Coverage.
27. Customer enters incident date, amount, description, severity, declaration, and files.
28. System validates coverage period, coverage limit, formats, size, count, and ownership.
29. System flags possible duplicate if detected.
30. System stores Claim and documents.
31. System notifies Claims Manager queue and Customer.

UC-CLM-002 Review and Decide Claim

Attribute	Value
Actor	Claims Manager
Goal	Reach final Claim outcome.
Preconditions	Approved Claims Manager and reviewable Claim.
Postconditions	Claim Approved or Rejected.

Main Flow

32. Claims Manager begins review.
33. Manager reviews Policy, Product, Coverage, Property, Claim, and evidence.
34. Manager may request evidence up to three times.
35. Manager records structured assessment values and Estimated Loss.
36. System calculates non-binding recommendation.
37. Manager records Approved or Rejected and mandatory reason.
38. System persists final state, audit, and Customer notification.

UC-REN-001 Renew Policy

Attribute	Value
Actor	Customer
Goal	Create a new one-year Policy from an eligible Policy.
Preconditions	Within renewal window; Policy not cancelled; no active renewal exists.
Postconditions	New Active Policy created after payment.

Main Flow

39. Customer opens eligible Policy and chooses Renew.
40. System creates renewal Quote with current data and configuration.

41. Customer may select eligible Product.
42. System recalculates Risk and Premium.
43. Customer completes payment.
44. System creates new Policy Number and renewal link.

8. Business Rules

8.1 Authentication and Authorization Rules

- Registration creates Customer or Pending Approval Claims Manager only.
- Exactly three roles exist: Customer, Claims Manager, Admin.
- Claims Manager cannot access review functions until approved and Active.
- Admin Claim-decision authority is denied.
- Email is unique username; phone is mandatory and not unique.
- Five failed logins lock the account for 15 minutes.
- Password change/reset invalidates all sessions.

8.2 Customer and Property Rules

- Customer may act only on owned records.
- Account holder/policy purchaser must be 18 or older.
- Property ownership transfer, deletion, and archive are outside scope.
- Property Value is 50,000-10,000,000 and Construction Year cannot be future.
- Risk-relevant Property changes invalidate affected active Quotes, not active Policy snapshots.

8.3 Product and Coverage Rules

- Only three approved customer-facing Products exist.
- One Product exists per Policy.
- Property Protection and Natural Disaster Protection may coexist for one Property.
- Comprehensive Protection cannot overlap either component Product on the same Property.
- Claims target a specific included Coverage.
- Inactive Product blocks new purchase and renewal but not existing Policy or Claim processing.

8.4 Quote and Pricing Rules

- Quote validity is 30 days.
- All Quotes remain in history.
- Multiple active Quotes are allowed.
- Existing Quote snapshots do not change when pricing changes.
- Base Rate, multipliers, charges, and limits are Admin-configurable and audited.

8.5 Policy and Payment Rules

- Policy begins Pending Payment and activates only after payment success.
- Effective Date is payment date; duration is one year.
- Backdating and manual expiry changes are prohibited.
- Pending Payment expires after 30 days.
- Payment method is simulated card; maximum five attempts.
- Cancellation is immediate, final, reason-required, and does not need Admin approval.

8.6 Renewal Rules

- Renewal window begins 30 days before expiry and ends 30 days after expiry.

- Cancelled Policy cannot renew.
- Only one active renewal process per Policy.
- Renewal recalculates risk and premium and creates a new Policy number.

8.7 Claim Rules

- Claim requires Active owned Policy, included Coverage, amount, incident date, description, severity, declaration, and evidence.
- Incident date must be within Policy period and not future.
- Amount cannot exceed coverage limit.
- Possible duplicates are flagged, not automatically rejected.
- At most three evidence requests; each deadline is 15 days.
- No automatic rejection after evidence deadline.
- Claims Manager is final decision maker; reason is mandatory.
- Approved and Rejected are final; rejected Claim may be resubmitted only as a new Claim.

8.8 Document Rules

- Supported formats are PDF, JPG, and PNG.
- Maximum size is 10 MB per file and maximum ten Claim files.
- Customer cannot delete evidence.
- Claims Manager may upload investigation documents.
- Downloads are authorized and audited.

8.9 Notification, Cache and Audit Rules

- Critical Policy, Claim, account, and Claims Manager approval notifications are mandatory.
- Email and SMS are not required for MVP.
- Products, Coverages, and reference data cache for 30 minutes.
- Admin changes immediately invalidate related cache entries.
- Critical business actions persist to AuditLog in addition to structured application logs.

9. User Journey

9.1 Customer Journey

Stage	Customer Action	System Action	Outcome
Register	Submit identity/contact/password	Validate, create Customer account, audit	Active account
Profile	Complete address and DOB	Validate ownership and fields	Profile complete
Property	Register vacation property	Validate risk inputs	Property available
Product	Review 3 products and included coverages	Return active cached catalogue	Product selected
Quote	Request Quote	Risk and premium calculation; snapshot	30-day Quote
Policy	Convert Quote	Create Pending Payment Policy	Payment required
Payment	Submit mock card details	Store attempt; activate on success	Active Policy
Claim	Select coverage; enter details; upload evidence	Validate, store, flag duplicate	Submitted Claim
Evidence	Respond to request	Store additional documents	Review resumes
Outcome	Open notification/Claim	Return final decision and reason	Outcome visible

Renew	Initiate within window	Reprice and create new Policy after payment	Renewed Active Policy

9.2 Claims Manager Journey

```
Register -> Upload Verification -> Pending Approval -> Approved -> Login -> Review Queue -> Begin Review -> Request Evidence if needed -> Record Assessment -> View Recommendation -> Final Decision
```

9.3 Admin Journey

```
Login -> Dashboard -> Approve/Reject Claims Manager -> Manage Account Status -> Manage Product Status -> Manage Pricing/Limits -> View Records -> Run/Export Reports
```

10. System Workflows

10.1 Customer Workflow

```
Register -> Login -> Complete Profile -> Register Property -> Select Product -> Generate Quote -> Create Pending-Payment Policy -> Simulated Payment -> Active Policy -> Submit Coverage-Specific Claim -> Track Outcome
```

10.2 Quote Workflow

```
Owned Property -> Active Product -> Eligibility -> Risk Score -> Risk Level -> Premium Breakdown -> Snapshot -> Active Quote (30 days)
```

10.3 Policy Workflow

```
Applicable Quote -> Pending Payment -> Payment Success -> Active -> Expired OR Customer Cancellation -> Cancelled
```

10.4 Renewal Workflow

```
Eligible Active/Recently Expired Policy -> Renewal Quote -> Recalculated Risk/Premium -> Payment -> New Active Policy
```

10.5 Claim Workflow

```
Active Policy -> Included Coverage -> Valid Claim + Evidence -> Submitted -> Under Review -> Optional Awaiting Documents -> Assessment -> Recommendation -> Approved/Rejected
```

10.6 Notification Workflow

```
Business Event -> Template Resolution -> Notification Record -> User Notification Center -> Deep Link -> Mark Read
```

10.7 Document Workflow

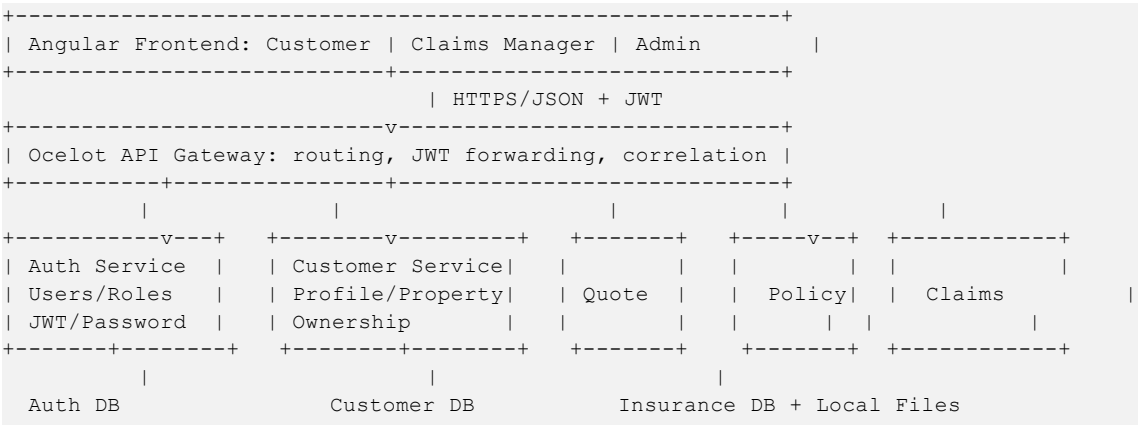
```
Authorized Upload -> Type/Size/Count Validation -> Metadata Persist -> File Write -> Audit -> Authorized Download -> Audit
```

10.8 Cache Workflow

```
Request -> In-Memory Key Lookup -> Hit Return / Miss DB Load -> TTL 30 min; Admin Change -> Immediate Invalidation
```

11. System Architecture

11.1 High-Level Architecture



11.2 Service Responsibilities

Component	Responsibilities
API Gateway	Route /api/v1 requests; forward JWT and correlation ID; controlled metadata logging; optional rate limiting.
Auth Service	Customer and CM registration, Admin approval status integration, login, JWT, password policy, lockout, reset/change password, roles.
Customer Service	Customer, CustomerProfile, VacationProperty, ownership validation, authoritative risk input data.
Quote & Underwriting	Products, Coverages, cache, pricing configuration, risk, premium, Quote snapshots.
Policy & Payment	Policy creation, payment attempts, activation, cancellation, expiry, renewal.
Claims	Claim submission, duplicate flagging, documents, evidence requests, assessment, recommendation, decision, notifications, reports.

11.3 Communication and Resilience

- Angular to Gateway and Gateway to Services use HTTPS REST/JSON.
- Insurance Service calls Customer Service synchronously for authoritative Property data.
- GET dependency calls use timeouts and limited safe retries.
- State-changing calls are not blindly retried.
- No distributed transaction; each service uses local transaction boundaries and idempotency/concurrency controls.
- Correlation IDs flow through all layers.

11.4 Deployment View

The MVP may run locally or in a controlled demo environment. A single SQL Server instance may host three independent databases. Service configuration, secrets, signing keys, and local storage roots are externalized from source control.

12. Domain Model

12.1 Entities

Bounded Context	Entities

Auth	User, Role, UserRole, PasswordResetToken, LoginAttempt
Customer	Customer, CustomerProfile, VacationProperty
Insurance Catalogue	InsuranceProduct, Coverage, ProductCoverage, PricingConfiguration
Quote & Underwriting	Quote, QuoteCoverage, RiskAssessment
Policy & Payment	Policy, PolicyCoverage, PaymentTransaction, RenewalLink
Claims	Claim, ClaimAssessment, EvidenceRequest, DocumentMetadata
Cross-Cutting	Notification, AuditLog

12.2 Aggregate Roots

User, Customer, VacationProperty, InsuranceProduct, Quote, Policy, Claim, Notification, and AuditLog are aggregate roots or independently controlled records. Child entities are changed only through aggregate use cases.

12.3 Relationships

```
User 1 ---- 0..1 Customer
Customer 1 ---- 1 CustomerProfile
Customer 1 ---- 0..* VacationProperty
InsuranceProduct * ---- * Coverage (ProductCoverage)
VacationProperty 1 ---- 0..* Quote
Quote 1 ---- 1 RiskAssessment
Quote 1 ---- 1..* QuoteCoverage
Quote 1 ---- 0..1 Policy
Policy 1 ---- 1..* PolicyCoverage
Policy 1 ---- 0..* PaymentTransaction
Policy 1 ---- 0..* Claim
Claim 1 ---- 1 ClaimAssessment (after review)
Claim 1 ---- 0..3 EvidenceRequest
Claim 1 ---- 1..10 DocumentMetadata
User 1 ---- 0..* Notification
```

12.4 Aggregate Invariants

- Quote controls Product, Coverage, Risk, Property, Pricing, and Premium snapshots.
- Policy controls issued Product/Coverage snapshots and lifecycle.
- Claim controls selected Coverage, evidence requests, assessment, recommendation, and final decision.
- No child record is independently modified through public CRUD endpoints.
- Cross-service identifiers are logical references; there are no cross-database foreign keys.

13. Database Design and Data Dictionary

13.1 Strategy

Use VacationHomeAuthDb, VacationHomeCustomerDb, and VacationHomeInsuranceDb. Each service owns its schema. Core insurance records are never hard-deleted. Soft delete is limited to eligible configuration/support records. Insurance records are retained for seven years from final activity; legal hold is outside MVP.

13.2 Auth Database Tables

Table	Key Fields / Constraints

Users	UserId PK; Email unique; PasswordHash; Status; FailedLoginCount; LockoutEnd; CreatedAt; UpdatedAt; RowVersion.
Roles	RoleId PK; Name unique; recognized roles only.
UserRoles	UserId + RoleId unique; one role per User in MVP.
PasswordResetTokens	TokenId PK; UserId; TokenHash; ExpiresAt; UsedAt.
LoginAttempts	AttemptId PK; UserId/email; Success; AttemptedAt; safe IP metadata.
AuditLogs	Auth-specific audit events where required.

13.3 Customer Database Tables

Table	Key Fields / Constraints
Customers	CustomerId PK; AuthUserId unique; FirstName; LastName; Phone; Status; CreatedAt; UpdatedAt.
CustomerProfiles	CustomerProfileId PK; CustomerId unique FK; DateOfBirth; address fields; CreatedAt; UpdatedAt.
VacationProperties	PropertyId PK; CustomerId FK; PropertyName; PropertyType enum; address; PropertyValue; ConstructionYear; OccupancyFrequency; HasSecuritySystem; PreviousClaimsCount; DisasterExposure; RowVersion.
AuditLogs	Profile/property change events.

13.4 Insurance Database Tables

Table	Key Fields / Constraints
InsuranceProducts	ProductId PK; ProductCode unique; ProductName; Description; IsActive; CreatedAt; UpdatedAt.
Coverages	CoverageId PK; CoverageCode unique; CoverageName; CoverageCharge; CoverageLimitPercentage; IsActive; UpdatedAt.
ProductCoverages	ProductId + CoverageId unique.
PricingConfigurations	PricingConfigurationId PK; BaseRate; Low/Medium/HighRiskMultiplier; EffectiveFrom; IsActive.
Quotes	QuoteId PK; QuoteNumber unique; CustomerId; PropertyId; ProductId; RiskScore; RiskLevel; premium breakdown; Status; ExpiresAt; snapshot JSON/columns; RowVersion.
QuoteCoverages	QuoteCoverageId PK; QuoteId FK; CoverageId; ChargeSnapshot; LimitSnapshot.
RiskAssessments	RiskAssessmentId PK; QuoteId unique FK; factor values; points; score; level; calculation version.
Policies	PolicyId PK; PolicyNumber unique; QuoteId unique; CustomerId; PropertyId; ProductId; premium; status; effective/expiry/cancel dates; cancellation reason; renewal source; RowVersion.
PolicyCoverages	PolicyCoverageId PK; PolicyId FK; CoverageId; ChargeSnapshot; LimitAmountSnapshot.
PaymentTransactions	PaymentId PK; PaymentNumber unique; PolicyId FK; CustomerId; Amount; Status; AttemptNumber; CreatedAt; no real card secret storage.
Claims	ClaimId PK; ClaimNumber unique; PolicyId FK; CustomerId; CoverageId; IncidentDate; ClaimAmount; Description; CustomerSeverity; Status; DuplicateFlag; SubmittedAt; Decision; DecisionReason; DecisionAt; RowVersion.
ClaimAssessments	AssessmentId PK; ClaimId unique; CoverageMatch; EvidenceQuality; VerifiedSeverity; IncidentValidation; EstimatedLoss; Score; Recommendation.
EvidenceRequests	EvidenceRequestId PK; ClaimId FK; Sequence 1-3; Reason; RequestedItems; RequestedAt; DueAt; RespondedAt; Status.

DocumentMetadata	DocumentId PK; ParentType; ParentId; FileName; StoredName; ContentType; SizeBytes; StoragePath; UploadedBy; UploadedAt.
Notifications	NotificationId PK; UserId; Title; Message; Type; RelatedEntityType/Id; IsRead; CreatedAt; ReadAt.
AuditLogs	AuditLogId PK; EntityName; EntityId; Action; OldValue; NewValue; PerformedBy; PerformedAt; IPAddress; CorrelationId.

13.5 Constraints and Indexes

- Unique indexes: Users(NormalizedEmail), Customers(AuthUserId), ProductCode, CoverageCode, QuoteNumber, PolicyNumber, ClaimNumber, PaymentNumber.
- Operational indexes: Properties(CustomerId), Quotes(CustomerId, CreatedAt), Quotes(PropertyId, Status), Policies(CustomerId, Status), Policies(PropertyId, Status), Claims(Status, SubmittedAt), Claims(PolicyId, IncidentDate), Notifications(UserId, IsRead, CreatedAt), AuditLogs(EntityName, EntityId, PerformedAt).
- Check constraints: positive amounts, Risk Score 0-100, Previous Claims 0-10, valid percentages, supported statuses, evidence request sequence 1-3, file size <=10 MB.
- Optimistic concurrency RowVersion is required for Policy and Claim transitions, product/pricing updates, and property edits.

13.6 Data Retention, Deletion and Recovery

Customer, Property, Quote, Policy, Payment, Claim, Claim Assessment, and Audit records are not deleted through business APIs. Eligible configuration/support records may use IsDeleted, DeletedAt, and DeletedBy. Retained records remain recoverable. Default retention is seven years; uploaded evidence follows the parent Claim retention.

14. API Specification Summary

All external APIs are versioned under /api/v1 and pass through Ocelot. Customer identity is derived from verified claims. Paths use lowercase nouns and explicit lifecycle actions.

Method	Path	Access
POST	/api/auth/customers/register	Public
POST	/api//auth/claims-managers/register	Public
POST	/api/auth/login	Public
POST	/api/auth/logout	Authenticated
POST	/api/auth/forgot-password	Public
POST	/api/auth/reset-password	Public
PUT	/api/auth/change-password	Authenticated
GET	/api/customers/profile	Customer
PUT	/api/customers/profile	Customer
POST	/api/properties	Customer
GET	/api/properties	Customer own
GET	/api/properties/{id}	Customer own
PUT	/api/properties/{id}	Customer own
GET	/api/products	Customer
GET	/api/products/{id}	Customer

POST	/api/quotes	Customer
GET	/api/quotes	Customer own
GET	/api/quotes/{id}	Customer own
POST	/api/policies	Customer
GET	/api/policies	Customer own
GET	/api/policies/{id}	Customer own
POST	/api/policies/{id}/cancel	Customer own
POST	/api/policies/{id}/renew	Customer own
POST	/api/payments	Customer own
GET	/api/payments/policy/{policyId}	Customer own
POST	/api/claims	Customer
GET	/api/claims	Customer own
GET	/api/claims/{id}	Customer own / authorized CM
POST	/api/claims/{id}/documents	Customer own / authorized CM
GET	/api/documents/{id}	Authorized parent access
GET	/api/claims-manager/claims	Claims Manager
PUT	/api/claims-manager/claims/{id}/review	Claims Manager
POST	/api/claims-manager/claims/{id}/evidence-requests	Claims Manager
PUT	/api/claims-manager/claims/{id}/assessment	Claims Manager
PUT	/api/claims-manager/claims/{id}/decision	Claims Manager
GET	/api/notifications	Authenticated
PUT	/api/notifications/{id}/read	Notification owner
GET	/api/admin/dashboard	Admin
GET	/api/admin/claims-managers/pending	Admin
PUT	/api/admin/claims-managers/{id}/approve	Admin
PUT	/api/admin/claims-managers/{id}/reject	Admin
PUT	/api/admin/users/{id}/status	Admin
PUT	/api/admin/products/{id}/status	Admin
PUT	/api/admin/pricing	Admin
PUT	/api/admin/coverages/{id}/limits	Admin
GET	/api/admin/reports/{type}	Admin
GET	/api/admin/reports/{type}/export	Admin

14.1 API Contract Principles

- POST lifecycle operations use idempotency where practical.
- List endpoints support page, pageSize (10/25/50), sort, search, and approved filters.
- CSV export uses the same approved filters as on-screen reports.
- File bytes are streamed through authorized APIs; storage paths are never public.
- Stable error codes and correlation IDs are returned.

- Cache headers may be used for safe catalogue responses, but server-side in-memory cache is authoritative for reference data.

15. Security Requirements

15.1 Authentication

- Signed JWT access tokens; services validate signature, issuer, audience, and expiration.
- Passwords use approved adaptive hashing and are never logged.
- Session timeout is 60 minutes; browser refresh may restore authenticated state.
- Five failed attempts cause a 15-minute lock.

15.2 Authorization

- Backend RBAC for Customer, Claims Manager, and Admin.
- JWT CustomerId supplied by client does not authorize access.
- Claims Manager access requires approved Active status.
- Admin has read-only insurance-record visibility and no Claim-decision authority.

15.3 Data Protection

- HTTPS outside local development.
- Secrets and signing keys outside source control.
- No real card data or CVV persistence.
- PII excluded from logs unless strictly required and authorized.
- EF Core parameterization and allow-listed enum/status values.

15.4 Document Security

- Validate MIME type, extension, size, count, parent permission, and file name.
- Generate safe stored names; prevent path traversal.
- Files are outside publicly served directories.
- Upload and download actions are audited.
- Virus scanning is recommended for production but not required in MVP.

15.5 Audit Security

- Audit records are append-oriented and not editable by business users.
- Old/new values must exclude passwords, JWTs, CVV, and document binary data.
- Audit searches are Admin-authorized.

15.6 Security Testing

- Cross-customer access, wrong-role access, tampered JWT, expired token, over-posting, injection, file-type spoofing, path traversal, duplicate decisions, account lockout, and session invalidation are tested.

16. Non-Functional Requirements

ID	Requirement
NFR-PERF-001	Core interactive APIs should target <=2 seconds under normal capstone demonstration load, excluding file transfer.
NFR-PERF-002	List APIs use asynchronous I/O and pagination.
NFR-PERF-003	Product, Coverage, pricing/reference data use 30-minute in-memory caching.

NFR-PERF-004	Admin configuration changes invalidate related cache immediately.
NFR-AVL-001	Core Customer and Claims Manager flows remain usable during normal demonstration and testing.
NFR-AVL-002	Services expose health endpoints.
NFR-REL-001	Unhandled failures return controlled responses and safe logs.
NFR-REL-002	Required service calls use timeouts and controlled handling.
NFR-REL-003	State transitions use optimistic concurrency.
NFR-REL-004	Duplicate/stale operations are detected and reported as conflicts.
NFR-MNT-001	Each service follows Clean Architecture and domain logic remains independent of controllers and EF Core.
NFR-MNT-002	Pricing, risk, limits, notification templates, and storage roots are configuration-driven within approved scope.
NFR-SCL-001	APIs are stateless except persistent business data and local in-memory cache.
NFR-SCL-002	Separate databases preserve service ownership; Redis is not required for the single-instance MVP.
NFR-SEC-001	Protected operations require authenticated, eligible, authorized actors.
NFR-SEC-002	Sensitive data is minimized in responses, logs, exports, and audit values.
NFR-SEC-003	Document files are never directly publicly addressable.
NFR-OBS-001	Important events and failures include correlation ID and safe context.
NFR-OBS-002	Business audit logs capture actor, action, entity, timestamp, and approved old/new values.
NFR-TST-001	Critical domain rules have unit tests; workflows have integration tests; core journeys have end-to-end tests.
NFR-USA-001	Core workflows are understandable without technical knowledge.
NFR-USA-002	Errors explain corrective action where practical.
NFR-USA-003	Customer, Claims Manager, and Admin navigation are clearly separated.
NFR-DATA-001	Core insurance records are retained seven years and are not hard deleted.
NFR-DATA-002	Quote and Policy snapshots remain historically immutable.

17. Validation and Error Handling

17.1 Validation Layers

Layer	Responsibilities
UI	Required fields, basic format/range feedback, disable obviously invalid submissions, file count/size guidance.
Application	Request shape, enum allow lists, role, ownership, dependency references, pagination limits.
Domain	Product overlap, Quote applicability, risk determinism, payment attempts, lifecycles, coverage inclusion, limits, evidence requests, final-state rules.
Database	PK/FK within database, unique/check constraints, RowVersion, positive values, status constraints.

17.2 Standard Error Response

```
{
  "success": false,
  "message": "The request could not be completed.",
  "errors": [{
   "code": "CLAIM_COVERAGE_NOT_INCLUDED",
```

```
    "field": "coverageId",
    "message": "The selected coverage is not included in this policy."
   }],
   "traceId": "correlation-value"
}
```

17.3 Status Standards

HTTP	Use
200	Successful retrieval or action
201	Resource created
400	Validation failure
401	Unauthenticated
403	Authenticated but unauthorized/ineligible
404	Unavailable or non-owned resource
409	Lifecycle, duplicate, idempotency, or concurrency conflict
413	File too large
415	Unsupported file format
429	Rate/attempt control where applied
503	Required dependency unavailable
500	Controlled unexpected failure

17.4 Stable Error Codes

- PRODUCT_INACTIVE
- PRODUCT_OVERLAP
- QUOTE_EXPIRED
- QUOTE_INVALIDATED
- QUOTE_ALREADY_CONVERTED
- POLICY_NOT_ACTIVE
- POLICY_RENEWAL_NOT_ALLOWED
- PAYMENT_ATTEMPTS_EXCEEDED
- CLAIM_COVERAGE_NOT_INCLUDED
- CLAIM_AMOUNT_EXCEEDS_LIMIT
- CLAIM_POSSIBLE_DUPLICATE
- EVIDENCE_REQUEST_LIMIT_REACHED
- DOCUMENT_FORMAT_UNSUPPORTED
- DOCUMENT_SIZE_EXCEEDED
- ACCOUNT_PENDING_APPROVAL
- ACCOUNT_LOCKED
- CONCURRENCY_CONFLICT

18. Testing Requirements

18.1 Unit Testing

- Risk scoring boundaries and score cap.
- Coverage-based pricing formula and snapshots.
- Product exclusivity and overlap checks.

- Quote expiry/invalidation and one-time conversion.
- Policy payment/activation/cancellation/renewal transitions.
- Payment attempt limit.
- Claim date, amount, included coverage, duplicate flagging, evidence request limit, recommendation score, and final-state rules.
- Cache key expiration and invalidation.

18.2 Integration Testing

- Customer and Claims Manager registration/onboarding.
- Admin approval/rejection and login eligibility.
- Customer Service Property lookup from Insurance Service.
- Product/coverage cache behavior.
- Quote persistence and Policy conversion.
- Payment attempt history and activation.
- Document metadata plus file persistence and authorized download.
- Notifications and AuditLog creation for each critical event.
- CSV report export.

18.3 End-to-End Workflows

- Customer: Register -> Profile -> Property -> Product -> Quote -> Payment -> Active Policy.
- Claim: Submit with evidence -> Review -> Evidence Request -> Response -> Assessment -> Recommendation -> Decision -> Notification.
- Claims Manager: Register -> Admin approval -> Login -> Review queue -> decision.
- Renewal: eligible Policy -> renewal Quote -> payment -> new Policy.
- Admin: Dashboard -> product deactivation -> immediate cache invalidation -> report export.

18.4 Example Test Catalogue

Test ID	Requirement	Scenario	Expected Result
TC-PROD-001	FR-PROD-002	Comprehensive requested while component policy active	409 overlap conflict
TC-QUOTE-001	FR-QUOTE-001	Valid Property and active Product	30-day Quote persisted
TC-QUOTE-002	FR-QUOTE-003	Property changed after Quote	Quote invalidated
TC-PAY-001	FR-PAY-001	Failed then successful mock payment	Attempts stored; Policy activates on success
TC-PAY-002	FR-PAY-001	Sixth attempt	Rejected after five attempts
TC-CLM-001	FR-CLM-001	Claim coverage included	Submitted
TC-CLM-002	FR-CLM-001	Amount over limit	Validation failure
TC-DOC-001	FR-DOC-001	11 MB PDF	413 rejected
TC-REV-001	FR-CLM-005	Assessment score 85	Recommend Approve
TC-REV-002	FR-CLM-006	Admin attempts Claim decision	403 denied
TC-CACHE-001	FR-CACHE-001	Admin deactivates Product	Cache invalidated immediately
TC-SEC-001	NFR-SEC-001	Customer requests another customer Policy	404/non-disclosing denial

19. Risk Analysis

Risk	Impact	Mitigation
Capstone rates mistaken for insurer rates	Misrepresentation	Label configuration as demonstration-only and snapshot calculation version.
In-memory cache staleness	Inactive Product briefly visible	Immediate invalidation and 30-minute TTL.
Local document storage loss	Evidence unavailable	Controlled path, backup guidance, metadata integrity, and production migration path.
Simulated payment mistaken for real payment	Security/compliance confusion	Explicit mock labels; store no real card secrets.
Claims recommendation treated as decision	Improper automation	UI and rules state recommendation is advisory; Claims Manager final action required.
Evidence deadline causes unfair rejection	Customer harm	No auto-reject; Claims Manager review required.
Pricing changes alter historical records	Audit inconsistency	Immutable Quote/Policy snapshots.
Optional email delays delivery	Scope risk	In-app notifications are mandatory; email stays future enhancement.
Document malware	Security exposure	Allow-listed types, size controls, safe names; production virus scan recommended.
One-week schedule	Incomplete core workflow	Prioritize approved vertical slice and exclude enterprise infrastructure.

20. Requirements Traceability Matrix

Requirement	User Story	Use Case/Flow	Entity	API	Verification
FR-AUTH-001	US-AUTH-001	UC-AUTH-001	User, Customer	POST /auth/customers/register	TC-AUTH-REG
FR-AUTH-002	US-CM-001	UC-CM-001	User, DocumentMetadata	POST /auth/claims-managers/register	TC-CM-REG
FR-PROP-001	US-PROP-001	Customer Journey	VacationProperty	POST /properties	TC-PROP-VALID
FR-PROD-001	US-PROD-001	Quote Workflow	Product, Coverage	GET /products	TC-CACHE-001
FR-RISK-001	US-QUOTE-001	UC-QUOTE-001	RiskAssessment	POST /quotes	TC-RISK-BOUNDS
FR-PREM-001	US-QUOTE-001	UC-QUOTE-001	Quote, PricingConfiguration	POST /quotes	TC-PREM-CALC
FR-POL-001	US-PAY-001	UC-POL-001	Policy	POST /policies	TC-POL-CREATE
FR-PAY-001	US-PAY-001	UC-POL-001	PaymentTransaction	POST /payments	TC-PAY-001/002
FR-POL-004	US-REN-001	UC-REN-001	Policy, RenewalLink	POST /policies/{id}/renew	TC-REN-WINDOW
FR-CLM-001	US-CLM-001	UC-CLM-001	Claim, DocumentMetadata	POST /claims	TC-CLM-001/002

FR-CLM-003	US-CM-002	UC-CLM-002	EvidenceRequest	POST /claims/{id}/evidence-requests	TC-EVID-LIMIT
FR-CLM-005	US-CM-002	UC-CLM-002	ClaimAssessment	PUT /claims/{id}/assessment	TC-REV-001
FR-CLM-006	US-CM-002	UC-CLM-002	Claim, ClaimAssessment	PUT /claims/{id}/decision	TC-REV-ROLE
FR-NTF-001	US-NTF-001	Notification Workflow	Notification	GET/PUT /notifications	TC-NTF-LINK
FR-ADM-004	US-ADM-001	Admin Journey	PricingConfiguration, Coverage	PUT /admin/pricing	TC-PRICE-SNAPSHOT
FR-RPT-001	US-RPT-001	Admin Journey	Read models	GET /admin/reports/*	TC-RPT-CSV
FR-CACHE-001	US-PROD-001	Cache Workflow	Product/Coverage cache	GET /products	TC-CACHE-001

20.2 Cross-Cutting Traceability

Requirement Group	Implementation	Verification
NFR-SEC-*	JWT, RBAC, ownership, document authorization, safe secrets	Security and API tests
NFR-REL-*	Transactions, RowVersion, idempotency, timeout handling	Conflict and rollback tests
NFR-OBS-*	Correlation IDs, structured logs, AuditLog	Audit inspection
NFR-DATA-*	Constraints, retention, immutable snapshots	Database/domain tests
NFR-PERF-*	Pagination, async I/O, in-memory cache	Performance and cache tests

21. Open Items, Assumptions and Decisions

21.1 Confirmed Decisions

- Three fixed customer-facing Products; Hurricane Coverage removed.
- In-app notifications required; real email and SMS excluded from MVP.
- Hybrid local document storage inside Insurance Service; no separate service.
- Simulated card payment; no real settlement.
- Manual renewal, no policy modification after activation other than cancellation/renewal.
- Claims Manager recommendation is advisory and human decision is final.
- Admin cannot decide Claims, create Products, delete Products, or edit insurance history.
- No Product versioning in MVP; historical Quote/Policy snapshots preserve issued terms.

21.2 Configurable Approved Values

- Base Rate and risk multipliers.
- Coverage charges and coverage-limit percentages.
- Property value bands and accepted property-value range.
- Notification message text and reference data labels.

21.3 Scope Exclusions

- Real payment gateway, refund, settlement, and PCI processing.
- Claim settlement/disbursement, fraud detection, appeals, reopening, and multi-level investigation.
- External insurer/property/risk integrations.

- Email/SMS provider integration.
- OCR, document versioning, Customer evidence deletion, public file URLs.
- Product builder, selectable riders, Coverage CRUD, Product versioning.
- Minor/guardian/trustee workflows.
- Redis, event bus, distributed transactions, Kubernetes, service mesh, and production cloud orchestration.

22. Success Criteria

22.1 Business Success Criteria

- Customer completes registration, profile, property, Quote, simulated payment, and Active Policy.
- Product selection and included coverages are correct and no legacy coverage options remain.
- Customer submits a valid coverage-specific Claim with mandatory evidence.
- Claims Manager is approved by Admin, completes assessment, sees recommendation, and records final decision.
- Customer receives and opens relevant notifications.
- Admin manages Product status, pricing, limits, users, dashboard, and CSV reports without altering Claim decisions.
- Renewal creates a new correctly priced Policy within the approved window.

22.2 Technical Success Criteria

- Angular works through Ocelot to three independent services and databases.
- JWT, RBAC, ownership, lockout, password reset/change, and session invalidation work.
- EF Core migrations, constraints, RowVersion, and snapshots work.
- Local hybrid document storage and authorized streaming work.
- Cache hit, expiry, and immediate invalidation work.
- Audit logs, correlation IDs, controlled errors, and health endpoints work.
- Automated tests cover critical domain logic and end-to-end workflows.

22.3 MVP Completion Gate

Must Be Complete	Must Not Block Completion
Customer vertical slice through Active Policy	Real email integration
Coverage-specific Claim through final decision	Real payment gateway
Claims Manager approval and review	OCR/virus scanning integration
Admin dashboard/configuration/reporting	Production orchestration
Notifications, documents, cache, audit	Advanced BI and analytics
Security, validation, persistence, tests	Selectable riders/product builder

23. Final SRS Summary

23.1 Business Scope

The system supports the approved vacation-home insurance lifecycle from user onboarding through Product selection, risk-based Quote, simulated payment, one-year Policy, manual renewal, coverage-specific Claim, evidence management, structured Claims Manager assessment, recommendation, and final decision.

23.2 System Scope

The solution contains Angular, Ocelot, Auth Service, Customer Service, Insurance Service, three SQL Server databases, local document storage, and in-memory cache. Notifications and document management remain internal capabilities, not separate services.

23.3 Product Baseline

Property Protection contains Dwelling, Other Structures, Personal Property, and Theft Protection. Natural Disaster Protection contains Flood, Earthquake, and Wildfire. Comprehensive Protection contains all seven coverages. Hurricane and all legacy customer-facing coverage choices are removed.

23.4 Implementation Readiness

This SRS is implementation-ready for Angular routes/components/guards/services, .NET APIs and domain logic, EF Core models/migrations, Ocelot routes, local file storage, cache keys/invalidation, automated testing, reporting, and demonstration planning. Any future feature outside the confirmed baseline requires formal change control.

Appendix A. Authorization Matrix

Action	Customer	Claims Manager	Admin
Register Customer	Yes	No	No
Register Claims Manager	No	Yes	No
Approve/Reject CM	No	No	Yes
Manage Own Profile	Yes	Yes	No
Create/Update Property	Own	No	No
Generate Quote	Own	No	No
Purchase/Cancel/Renew Policy	Own	No	No
Submit Claim	Own	No	No
Review/Decide Claim	No	Yes	No
Request Evidence	No	Yes	No
View All Customers/Policies/Claims/Payments	No	No	Read-only
Activate/Deactivate Accounts	No	No	Yes
Activate/Deactivate Products	No	No	Yes
Manage Pricing/Limits	No	No	Yes
Export Reports	No	No	Yes

Appendix B. Audit Event Catalogue

Domain	Audited Events
Authentication	Registration; login success/failure; lockout; logout; reset/change password
Customer	Profile update; Property create/update
Quote/Policy	Quote generation/invalidation/conversion; Policy create/activate/cancel/expire/renew
Payment	Every attempt; success; failure; attempt limit reached
Claims	Submission; duplicate flag; review start; evidence request/response; assessment; recommendation; final decision
Documents	Upload and download

Administration	CM approve/reject/reapply; account status; Product status; pricing/limit change; report export
Cache	Configuration change and cache invalidation failure

Appendix C. Notification Catalogue

Recipient/Domain	Mandatory In-App Events
Customer	Registration successful; password reset/change; account deactivated/reactivated
Policy	Quote generated/expiring; payment success/failure; Policy activated/cancelled/expiring/expired; renewal available/completed
Claim Customer	Claim submitted; under review; evidence requested; evidence deadline expired; approved; rejected
Claims Manager	Registration submitted; approved; rejected; new Claim available; evidence response received; decision SLA reminder
Admin	New CM registration; significant configuration/audit alert where implemented

Notification records include delivery status Created, read status, deep link, and history. Retry applies only to transient notification-record creation failures within the service transaction/recovery process. External email delivery retry is outside MVP.

Appendix D. Legacy Product Migration Matrix

Legacy Item	Revised Treatment
Fire	Removed as standalone; property fire-type damage handled under applicable Dwelling/Other Structures coverage rules defined by the product.
Theft	Mapped to Theft Protection under Property Protection and Comprehensive Protection.
Natural Disaster	Replaced by Natural Disaster Protection Product with Flood, Earthquake, and Wildfire coverages.
Personal Liability	Removed from approved scope.
Personal Belongings	Mapped to Personal Property Coverage.
Hurricane	Removed from revised Product catalogue.
Coverage selection UI	Replaced by Product selection with visible included coverages.
GET /coverages customer entry	Replaced by GET /products and Product details; Coverage remains internal/visible within Product.
Quote/Policy join tables	Retained as snapshot structures but populated from selected Product bundle.
Existing data migration	Seed three Products and seven Coverages; map supported legacy records where deterministic; unsupported legacy values require controlled migration exception and must not be silently remapped.

Appendix E. Report Data Definitions

Report	Columns
Customer	Customer ID, name, email, phone, status, registered date, property count, policy count, claim count, last login.
Property	Property ID/name/type/location/value/construction year/occupancy/security/disaster exposure/customer.

Policy	Policy number, customer, property, product, premium, status, effective date, expiry date, created date.
Claim	Claim number, policy number, coverage, customer, amount, recommendation score, status, decision, decision date, Claims Manager.
Payment	Payment number, policy number, customer, amount, status, attempt number, transaction date.
Product	Product code/name/status, included coverages, coverage charges/limits, last updated date.

Appendix F. UI Screen Inventory

Area	Screens
Customer	Register; Login; Forgot/Reset Password; Dashboard; Profile; Properties; Product Catalogue; Quote Generate/Detail/History; Policy Detail; Mock Payment; Renewal; Claim Submit/Detail; Evidence Upload; Notifications.
Claims Manager	Register; Pending Approval; Login; Dashboard; Review Queue; Claim Detail; Evidence Request; Assessment; Recommendation; Decision; Notifications; Profile.
Admin	Login; Dashboard; Pending CM Approvals; User Status; Product Status; Pricing/Limits; Customer/Property/Policy/Claim/Payment read-only views; Reports and CSV Export; Audit view where implemented.